

Proof of the Hilbert sequence space l^2 being a metric space

Definition of metric space: A *metric space* is a pair (X, d) , where X is a set and d is a *metric on X* (or *distance function on X*), that is, a function defined on $X \times X$ such that for all $x, y, z \in X$ we have:

- (M1) d is real-valued, finite and nonnegative.
- (M2) $d(x, y) = 0$ if and only if $x = y$.
- (M3) $d(x, y) = d(y, x)$.
- (M4) $d(x, y) \leq d(x, z) + d(z, y)$

Let $p \geq 1$ be a fixed real number. By definition, each element in the space l^p is a sequence $x = (\xi_j) = (\xi_1, \xi_2, \dots)$ of numbers such that $|\xi_1|^p + |\xi_2|^p + \dots$ converges, meaning:

$$\sum_{j=1}^{\infty} |\xi_j|^p < \infty \quad (p \geq 1, \text{ fixed}) \quad (1)$$

and the metric is defined by

$$d(x, y) = \left(\sum_{j=1}^{\infty} |\xi_j - \eta_j|^p \right)^{1/p} \quad (2)$$

where $y = (\eta_j)$ and $\sum |\eta_j|^p < \infty$.

In the case $p = 2$ we have the famous *Hilbert sequence space* l^2 with metric defined by

$$d(x, y) = \sqrt{\sum_{j=1}^{\infty} |\xi_j - \eta_j|^2}. \quad (3)$$

This space is the earliest example of what is now called a *Hilbert space*.

Here, we prove that l^p is a metric space. Clearly, (2) satisfies M1 to M3 provided the series on the right converges. We shall prove that it does converge and that M4 is satisfied. Proceeding stepwise, we shall derive

- (a) an auxiliary inequality,
- (b) the Hölder inequality from (a),
- (c) the Minkowski inequality from (b),
- (d) the triangle inequality (M4) from (c).

(a) Let $p > 1$ and define q by

$$\frac{1}{p} + \frac{1}{q} = 1. \quad (4)$$

p and q are then called **conjugate exponents**. From (4) we have

$$1 = \frac{p+q}{pq}, \quad pq = p+q, \quad (p-1)(q-1) = 1. \quad (5)$$

Hence $1/(p-1) = q-1$, so that

$$u = t^{p-1} \quad \text{implies} \quad t = u^{q-1}.$$

Let α and β be any positive numbers. Consider them as numbers at the two axis t and u . Notice in Figure 1, no matter how α and β were changed, the area of the integral exceeds the area of the triangle. Thus, we obtain by integration the inequality

$$\alpha\beta \leq \int_0^\alpha t^{p-1} dt + \int_0^\beta u^{q-1} du = \frac{\alpha^p}{p} + \frac{\beta^q}{q}. \quad (6)$$

Note that this inequality is trivially true if $\alpha = 0$ or $\beta = 0$.

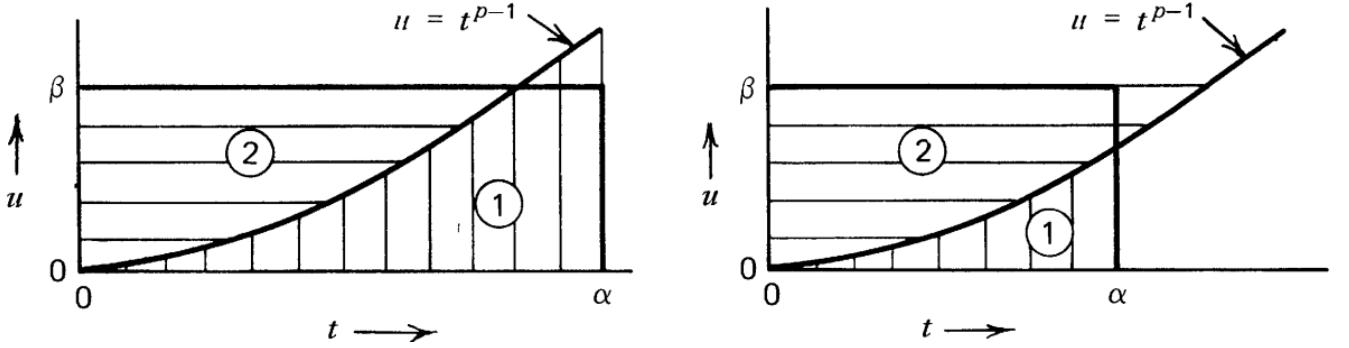


Figure 1: Inequality (6), where ① corresponds to the first integral in (6) and ② to the second

(b) Let $(\tilde{\xi}_j)$ and $(\tilde{\eta}_j)$ be such that

$$\sum |\tilde{\xi}_j|^p = 1, \quad \sum |\tilde{\eta}_j|^q = 1. \quad (7)$$

Setting $\alpha = |\tilde{\xi}_j|$ and $\beta = |\tilde{\eta}_j|$, we have from (6) the inequality

$$|\tilde{\xi}_j \tilde{\eta}_j| \leq \frac{1}{p} |\tilde{\xi}_j|^p + \frac{1}{q} |\tilde{\eta}_j|^q.$$

If we sum over j and use (7) and (4), we obtain

$$\sum |\tilde{\xi}_j \tilde{\eta}_j| \leq \frac{1}{p} + \frac{1}{q} = 1. \quad (8)$$

We now take any nonzero $x = (\xi_j) \in l^p$ and $y = (\eta_j) \in l^q$ and set

$$\tilde{\xi}_j = \frac{\xi_j}{(\sum |\xi_k|^p)^{1/p}}, \quad \tilde{\eta}_j = \frac{\eta_j}{(\sum |\eta_m|^q)^{1/q}}. \quad (9)$$

Then (7) is satisfied, so that we may apply (8). Substituting (9) into (8) and multiplying the resulting inequality by the product of the denominators in (9), we arrive at the **Hölder inequality for sums**

$$\sum_{j=1}^{\infty} |\xi_j \eta_j| \leq \left(\sum_{k=1}^{\infty} |\xi_k|^p \right)^{1/p} \left(\sum_{m=1}^{\infty} |\eta_m|^q \right)^{1/q} \quad (10)$$

where $p > 1$ and $1/p + 1/q = 1$. This inequality was given by O. Hölder.

If $p = 2$, then $q = 2$ and (10) yields the **Cauchy-Schwarz inequality for sums**

$$\sum_{j=1}^{\infty} |\xi_j \eta_j| \leq \sqrt{\sum_{k=1}^{\infty} |\xi_k|^2} \sqrt{\sum_{m=1}^{\infty} |\eta_m|^2}. \quad (11)$$

(c) We now prove the **Minkowski inequality for sums**

$$\left(\sum_{j=1}^{\infty} |\xi_j + \eta_j|^p \right)^{1/p} \leq \left(\sum_{k=1}^{\infty} |\xi_k|^p \right)^{1/p} + \left(\sum_{m=1}^{\infty} |\eta_m|^p \right)^{1/p} \quad (12)$$

where $x = (\xi_j) \in l^p$ and $y = (\eta_j) \in l^p$, and $p \geq 1$. For finite sums this inequality was given by H. Minkowski.

For $p = 1$ the inequality follows from the triangle inequality for numbers. Let $p > 1$. To simplify the formulas we shall write $\xi_j + \eta_j = \omega_j$. The triangle inequality for numbers gives

$$\begin{aligned} |\omega_j|^p &= |\xi_j + \eta_j| |\omega_j|^{p-1} \\ &\leq (|\xi_j| + |\eta_j|) |\omega_j|^{p-1}. \end{aligned}$$

Summing over j from 1 to any fixed n , we obtain

$$\sum |\omega_j|^p \leq \sum |\xi_j| |\omega_j|^{p-1} + \sum |\eta_j| |\omega_j|^{p-1}. \quad (13)$$

To the first sum on the right we apply the Hölder inequality, finding

$$\sum |\xi_j| |\omega_j|^{p-1} \leq \left[\sum |\xi_k|^p \right]^{1/p} \left[\sum (|\omega_m|^{p-1})^q \right]^{1/q}.$$

Notice that

$$(p-1)q = p$$

because $pq = p + q$ (see (5)). Treating the last sum in (13) in a similar fashion, we obtain

$$\sum |\eta_j| |\omega_j|^{p-1} \leq \left[\sum |\eta_k|^p \right]^{1/p} \left[\sum |\omega_m|^p \right]^{1/q}.$$

Together,

$$\sum |\omega_j|^p \leq \left\{ \left[\sum |\xi_k|^p \right]^{1/p} + \left[\sum |\eta_k|^p \right]^{1/p} \right\} \left(\sum |\omega_m|^p \right)^{1/q}.$$

Dividing by the last factor on the right and noting that $1 - 1/q = 1/p$, we obtain (12) with n instead of ∞ . We now let $n \rightarrow \infty$. On the right this yields two series which converge because $x, y \in l^p$. Hence the series on the left also converges, and (12) is proved.

(d) From (12) it follows that for x and y in l^p the series in (2) converges. (12) also yields the triangle inequality. In fact, taking any $x, y, z \in l^p$, writing $z = (\zeta_j)$ and using the triangle inequality for numbers and then (12), we obtain

$$\begin{aligned} d(x, y) &= \left(\sum |\xi_j - \eta_j|^p \right)^{1/p} \\ &\leq \left(\sum [|\xi_j - \zeta_j| + |\zeta_j - \eta_j|]^p \right)^{1/p} \\ &\leq \left(\sum |\xi_j - \zeta_j|^p \right)^{1/p} + \left(\sum |\zeta_j - \eta_j|^p \right)^{1/p} \\ &= d(x, z) + d(z, y). \end{aligned}$$

This completes the proof that l^p is a metric space. $\square$